

Survey Workbench v2.1

User Manual

June 2026

Version 2.1 – Internal Use Only

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1 Overview

The Survey Workbench is a comprehensive tool for managing participant folders and extracting survey data from questionnaires. Version 2.1 introduces field mapping auto-generation from form templates, non-participant-specific column naming for multi-participant files, and streamlined configuration management.

1.1 Key Features

- Dynamic questionnaire configuration (unlimited types)
- Automated participant folder generation
- Data extraction from completed questionnaires to Excel
- Configuration save/load system
- Individual questionnaire customization
- Memory-optimized processing

1.2 System Requirements

- Windows 10/11
- Microsoft Excel (for data extraction)
- Sufficient disk space for participant folders

2 Getting Started

When you first launch Survey Workbench v2.1, you'll see the main window with two primary sections:

1. **Generate Participant Folder** – Creates organized folders with questionnaires
2. **Extract Data** – Reads completed questionnaires and appends to masterfile (CSV, XLS, or XLSX)

3 Generate Participant Folder

This section allows you to create standardized participant folders with multiple questionnaire types.

3.1 Step-by-Step Guide

3.1.1 Step 1: Enter Participant ID

Enter a unique identifier for the participant (e.g., "WaS3_001").

- This ID will be used in folder and file names
- Use consistent naming across all participants
- The field cannot be empty

3.1.2 Step 2: Select Target Folder

1. Click **Search** next to “Target folder PATH”
2. Choose where participant folders will be created
3. Example: C:\Studies\WaS3\Participants

3.1.3 Step 3: Configure Questionnaire Types

1. Enter the number of different questionnaire types needed
2. Click **Confirm** to create configuration rows

A scrollable area will appear with configuration fields for each questionnaire type.

3.1.4 Step 4: Configure Each Questionnaire

For each questionnaire type, you’ll configure three parameters:

Survey Name:

- Enter the questionnaire identifier (e.g., “demographics”, “nasa1”, “comments”)
- This name will be part of the output filename
- Use descriptive, consistent names

Template PATH:

- Click **Search** to select the PDF template file
- Example: C:\Templates\WaS3_demographics.pdf
- Template must be a valid PDF file

Number of copies:

- Enter how many copies of this questionnaire to create
- Default is 1
- Use multiple copies for repeated measures (e.g., 8 copies for NASA-TLX)

3.1.5 Step 5: Generate Folder

Click **Generate Participant Folder**. The system will:

1. Create a folder named with the Participant ID
2. Copy all questionnaires with standardized naming
3. Open the **Prefill Dialog** (see below)

3.1.6 Step 6: Prefill Dialog

After generating, a table appears showing all PDF form fields grouped for readability:

- **Field** – the system field name (e.g., `Check1`, `Text2`)
- **Name** – the user-friendly label from the field mapping (e.g., “Gender”)
- **Data** – an editable value to prefill into the PDF before handover

Checkbox grouping: Checkboxes sharing a base name (e.g., `Check1_1`, `Check1_2`, `Check1_3`) are shown as a single row `Check1`. Enter the option number to prefill (e.g., 1 to check option 1, leaving the rest unchecked). The system translates this to `/Yes` and `/Off` PDF values automatically.

Name column: Editing the Name column here updates the field mapping for the active configuration, which is also used as the column header in the masterfile.

3.2 File Naming Convention

- Single copy: `{ParticipantID}_{SurveyName}.pdf`
- Multiple copies: `{ParticipantID}_{SurveyName}1.pdf`, `{ParticipantID}_{SurveyName}2.pdf`, etc.

3.3 Example Output

```
Exp_001/
├── Exp_001_demographics.pdf
├── Exp_001_quest1.pdf
├── Exp_001_quest2.pdf
├── Exp_001_quest3.pdf
├── Exp_001_quest4.pdf
├── Exp_001_quest5.pdf
├── Exp_001_quest6.pdf
├── Exp_001_quest7.pdf
├── Exp_001_quest8.pdf
└── Exp_001_comments.pdf
```

4 Extract Data

This section reads completed questionnaire data and appends it to a masterfile. The system automatically detects the format (CSV, XLS, or XLSX) from your selected file and uses the appropriate method.

4.1 Prerequisites

- Completed questionnaires must have “_Extract Data.csv” files
- CSV files must be in the participant folder
- A masterfile must exist (.csv, .xls, or .xlsx format)
- The masterfile will be updated with new participant data

4.2 Step-by-Step Guide

4.2.1 Step 1: Select Source Folder

1. Click **Search** next to “Source folder PATH”
2. Choose the folder containing participant folders
3. Example: C:\Studies\WaS3\Completed

4.2.2 Step 2: Select Masterfile

1. Click **Search** next to “Masterfile PATH”
2. Choose your masterfile (.csv, .xls, or .xlsx)
3. The system will auto-detect the format and use the appropriate extraction method

4.2.3 Step 3: Enter Participant ID

- Enter the ID of the participant whose data you want to extract
- Must match the folder name exactly
- Case-sensitive

4.2.4 Step 4: Extract Data

Click **Extract Data to Masterfile**. The system will automatically detect your file format and proceed accordingly:

For CSV files (.csv)

1. Find all *_Extract Data.csv files in the participant folder
2. Read data from each CSV file
3. Append a new row to the CSV masterfile
4. Merge column headers if new fields appear
5. Save the updated file

For Excel files (.xls or .xlsx)

1. Find all *_Extract Data.csv files in the participant folder
2. Read data from each CSV file
3. Open the Excel masterfile
4. Find the next empty row
5. Write participant ID and all extracted data
6. Create column headers if this is the first row
7. Save the file (XLS files stay as .xls, XLSX stay as .xlsx)

4.3 Choosing Masterfile Format

Use **CSV (.csv)** when:

- You want a universal, plain-text format
- You plan to import data into R, Python, SPSS, or other analysis tools
- You need maximum compatibility across platforms and software
- You want human-readable, version-control-friendly files
- File size is a concern (CSV is smaller)

Use **Excel (.xls or .xlsx)** when:

- You want to work directly in Excel with formulas/pivot tables
- You need Excel-specific features (formatting, charts, multiple sheets)
- Your workflow is primarily in Microsoft Excel
- You're using older Excel versions (use .xls for Excel 2003 and earlier)
- You're using newer Excel versions (use .xlsx for Excel 2007+)

All formats:

- Append new participants as rows to the masterfile
- Automatically merge column headers when new fields appear
- Preserve all previous data in the masterfile
- The tool auto-detects the format – just select your file

4.4 Data Organization

- Participant ID is written in the first column
- Each field gets a column named with the field identifier (e.g., **Age**, **Mental_Demand**), or its mapped friendly name if a field mapping is configured
- Column names are *non-participant-specific*, allowing multiple participants to be added to the same masterfile
- Checkbox groups (e.g., **Check1_1**, **Check1_2**) are collapsed into a single column (e.g., **Check1**) containing the option number that was checked
- Example columns: **Age**, **Mental_Demand**, **Gender** (if **Check1** is mapped to “Gender”)

4.4.1 Step 5: Verify Results

- A success message shows how many CSV files were processed
- Check the Excel file to verify data was written correctly

5 Configuration Management

Save and load your questionnaire configurations for repeated use.

5.1 Save Configuration

1. Set up your questionnaires in the Generate section
2. Go to **File** → **Save Configuration**
3. Enter a descriptive configuration name
4. Click **Save configuration**

5.1.1 What's Saved

- Target folder path (for generation)
- Number of questionnaire types
- All questionnaire names, template paths, and copy counts
- Source folder path (for extraction)
- Masterfile path (for extraction)
- **Field mapping** (system field identifiers → user-friendly column names) – auto-generated from templates and editable via *Save/Edit Field Mapping*

5.2 Field Mapping

Field mapping defines user-friendly names for system field identifiers (e.g., mapping **Check1** to “Gender”). These names appear in the prefill dialog and as column headers in the masterfile.

5.2.1 Auto-Generation

When you click **File** → **Save/Edit Field Mapping**, the system automatically scans your configured form templates and generates an initial field mapping. This ensures the mapping always matches the actual form fields in your templates.

5.2.2 Editing the Mapping

1. Go to **File** → **Save/Edit Field Mapping**
2. A table shows all fields (**Field** column, read-only) auto-detected from your form templates
3. Edit the **Name** column to set user-friendly labels
4. Click **Save** – the mapping is saved to the active configuration

5.2.3 Persistence

- The mapping is automatically saved with your project configuration
- When you load a configuration, its field mapping loads automatically
- The mapping is applied when extracting data and during the prefill dialog
- No separate load/delete operations needed – field mappings are integral to the project configuration

5.2.4 Overwriting Configurations

If a configuration with the same name exists, you'll be asked to confirm:

- Choose **Yes** to replace the old configuration
- Choose **No** to cancel

5.3 Load Configuration

1. Go to **File** → **Load/Delete Configuration**
2. Select a configuration from the dropdown
3. Click **Load configuration**

All saved settings will be restored to the interface.

5.4 Delete Configuration

1. Go to **File** → **Load/Delete Configuration**
2. Select a configuration from the dropdown
3. Click **Delete configuration**
4. Confirm the deletion

6 Troubleshooting

6.1 Common Error Messages

6.1.1 “Please enter a participant ID!”

- Make sure you've entered an ID in the Participant ID field
- The field cannot be empty

6.1.2 “Please select a target folder!”

- Click Search and choose a valid folder
- The folder must exist and be accessible

6.1.3 “Please configure questionnaires!”

- Enter a number in “Number of questionnaire types”
- Click Confirm to create configuration rows

6.1.4 “Participant folder not found”

- Verify the Source folder is correct
- Check that the Participant ID matches the folder name exactly
- Folder names are case-sensitive

6.1.5 “No Extract Data CSV files found”

- Check that questionnaires have been processed
- CSV files must end with “_Extract Data.csv”
- Verify files are in the participant folder

6.1.6 “Folder already exists. Overwrite?”

- Choose **Yes** to replace existing files
- Choose **No** to cancel and preserve existing data

6.2 Excel-Related Issues

- Ensure Excel is installed and accessible
- Close the Excel file if it’s already open
- Check file permissions
- Verify the Excel file is not corrupted

6.3 Best Practices

6.3.1 Use Consistent Naming

- Use the same participant ID format across all participants
- Example: WaS3_001, WaS3_002, WaS3_003, etc.
- Avoid special characters in IDs

6.3.2 Organize Templates

- Keep all questionnaire templates in one folder
- Use descriptive filenames
- Maintain backups of template files

6.3.3 Save Configurations

- Create configurations for each study type
- Use descriptive configuration names
- Update configurations when templates change

6.3.4 Backup Data

- Regularly backup your Excel data files
- Keep copies of completed questionnaires
- Use version control for configuration files

6.3.5 Test First

- Test with a sample participant before processing real data
- Verify Excel output is formatted correctly
- Check all questionnaire templates load correctly

7 Advanced Features

7.1 Multiple Questionnaire Copies

When you need repeated measures:

1. Set “Number of copies” to the desired amount
2. Files will be numbered sequentially: `name1.pdf`, `name2.pdf`, etc.
3. Useful for: NASA-TLX repeated measures, multiple trials, time-series data

7.2 Custom Excel Sheets

The extraction looks for a sheet named “Data”:

- If found, data is written to this sheet
- If not found, data is written to the first sheet
- Headers are created automatically on the first row
- Existing headers are preserved and matched

7.3 File Column Filtering

The extraction automatically skips any column named “File” from CSV data to avoid redundant path information.

7.4 Form Field Naming Convention

For consistent CSV extraction, name PDF form fields using this pattern: `NameX(_N)(_n)`

- **Name**: Use **Text** for text fields and **Check** for checkboxes.
- **X**: Field group number. Fields that belong to the same question share the same **X**.
- **_N**: Checkbox option number (only for **Check** fields).
- For binary checkboxes, use `_1` for Yes and `_0` for No.
- For multi-option checkboxes, number options in ascending order: `_0`, `_1`, `_2`, etc.
- **_n**: Append this suffix for fields that should not be preferred for extraction.

Examples: `Text1`, `Check4_1`, `Check4_0`, `Text11_n`

8 Technical Information

8.1 Version History

8.1.1 v2.1 (June 2026)

- Field mapping auto-generated from form templates
- Non-participant-specific column names (support multiple participants per masterfile)
- Streamlined field mapping workflow (integrated with project configuration)
- Removed redundant load/delete field mapping operations
- Enhanced type hints and code quality
- Updated documentation for v2.1 features

8.1.2 v2.0 (February 2026)

- Dynamic questionnaire support (unlimited types)
- Individual configuration per questionnaire
- Improved data extraction algorithm
- Enhanced error handling
- Configuration save/load system
- Memory-optimized processing
- Comprehensive type hints

8.1.3 v1.2 (Previous)

- Fixed questionnaire count
- Basic data extraction
- Simple configuration

8.2 Credits

Version: 2.1
Release Date: June 2026

8.3 Technology Stack

- Python 3.13.7
- PyQt5 5.15.11 (GUI framework)
- xlwings 0.33.20 (Excel integration)
- PyInstaller 6.14.2 (Executable packaging)